

reverseDigits

Write a C function that takes in a positive integer number, reverses its digits and returns the result to the calling function. You may assume that the last digit of the input number is not 0, i.e. the number will not be in the form of 1110, 1200, etc. Write two iterative versions of the function. The function **reverseDigits1()** returns the computed result, while **reverseDigits2()** passes the result through the pointer parameter result. The function prototypes are given as follows:

```
int reverseDigits1(int num);  
void reverseDigits2(int num, int *result);
```

A sample program template is given below to test the functions:

```
#include <stdio.h>  
int reverseDigits1(int num);  
void reverseDigits2(int num, int *result);  
int main()  
{  
    int num, result=999;  
  
    printf("Enter a number: \n");  
    scanf("%d", &num);  
    printf("reverseDigits1(): %d\n", reverseDigits1(num));  
    reverseDigits2(num, &result);  
    printf("reverseDigits2(): %d\n", result);  
    return 0;  
}  
int reverseDigits1(int num)  
{  
    /* Write your code here */  
}  
void reverseDigits2(int num, int *result)  
{  
    /* Write your code here */  
}
```

Some sample input and output sessions are given below:

(1) Test Case 1:

Enter a number:

12045

reverseDigits1(): 54021

reverseDigits2(): 54021

(2) Test Case 2:

Enter a number:

123

reverseDigits1(): 321

reverseDigits2(): 321

(3) Test Case 3:
Enter a number:
8
reverseDigits1(): 8
reverseDigits2(): 8